

# **Whole-MHC pangenome graph of 754 haplotypes: where it is and how to map reads to it**

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## One graph holds the MHC of 754 haplotypes from five pangenome projects

- ▶ Haplotypes: APR (UAE Arab) 106, HPRC release 2 464, JaSaPaGe (Saudi, Japanese) 38, K-PanRef (Korean) 28, CPC (Chinese) 116, GRCh38, CHM13
- ▶ Input: the MHC segment cut from each haplotype by our HLA workflow (GRCh38 chr6:28.4–33.6 Mb with flanks)
- ▶ Built with Minigraph-Cactus (Cactus 3.3.0) on the NIG supercomputer: 3 h 43 min, 32 cores
- ▶ Reference path GRCh38#0#ctg1 = GRCh38 chr6:28,410,701–33,580,488 (graph position + 28,410,700 = chr6 coordinate)
- ▶ Graph: 417,896 nodes, 579,137 edges, 5.9 Mb of sequence

## All files are in one team folder on the NIG supercomputer

`/home/asianhla/data/upload/HLA/mhc_graph/`

- ▶ Mapping: `MHC.gbz` (graph with haplotypes, clipped), `MHC.dist`, `MHC.shortread.withzip.min`, `MHC.shortread.zipcodes` (vg giraffe indexes)
- ▶ Other formats: `MHC.gfa.gz`, `MHC.full.gbz` / `.full.gfa.gz` / `.full.og` (unclipped), `MHC.full.hal` (Cactus alignment)
- ▶ Variants: `MHC.vcf.gz` against GRCh38 (reference path `ctg1`)
- ▶ `contig_names.tsv`: renamed contigs (`ctg<N>`) back to original names and coordinates; `README.md`: details
- ▶ Reads to map:  
`/home/asianhla/data/upload/1000G_MHC/cram/<sample>.mhc.cram`, 2,504 1000 Genomes samples (in progress)
- ▶ `vg`: `/home/leechuck/hla/cactus/cactus-bin-v3.3.0/bin/vg` (version 1.76.1)

## Two commands map a 1000 Genomes sample to the graph (GAF or BAM)

```
# run in a Slurm job (sbatch -c 8 --mem=48G); this reference copy is readable from all nodes
REF=/home/leechuck/hla/ref1kg/local/GRCh38_full_analysis_set_plus_decoy_hla.fa
VG=/home/leechuck/hla/cactus/cactus-bin-v3.3.0/bin/vg
G=/home/asianhla/data/upload/HLA/mhc_graph
IDX="-Z $G/MHC.gbz -d $G/MHC.dist \
    -m $G/MHC.shortread.withzip.min -z $G/MHC.shortread.zipcodes"
CRAM=/home/asianhla/data/upload/1000G_MHC/cram/HG00096.mhc.cram

# 1. reads from the MHC CRAM
samtools collate -u -O --reference $REF $CRAM tmp \
| samtools fastq -F 0x900 --reference $REF -1 r1.fq.gz -2 r2.fq.gz \
    -O /dev/null -s /dev/null -n -

# 2a. align to the graph, graph coordinates (GAF)
$VG giraffe -t 8 $IDX -f r1.fq.gz -f r2.fq.gz -o gaf > HG00096.gaf

# 2b. or BAM on contig GRCh38#0#ctg1 (chr6 position = position + 28,410,700)
$VG giraffe -t 8 $IDX -f r1.fq.gz -f r2.fq.gz -o BAM | samtools sort -o HG00096.graph.bam -
```

## Test: 99.5% of HG00096 MHC read pairs align in 26 seconds

- ▶ Input: all 657,052 read pairs that bwa had placed in the MHC, chr6 alternative haplotypes or HLA contigs (1000G high-coverage CRAM of HG00096)
- ▶ vg giraffe, 8 threads: 26 s, 0.7 GB memory
- ▶ 654,059 pairs (99.5%) aligned; 625,590 (95.2%) with  $\text{MAPQ} \geq 20$
- ▶ Script: `mhc_graph/giraffe_test.sh`

## Four points to check before using the graph

- ▶ The graph covers only the MHC: reads from HLA-like sequence elsewhere in the genome can be forced onto it; start from reads already placed in the MHC
- ▶ giraffe refuses to start if MHC.dist is newer than the minimizer index (after copying files): `touch MHC.shortread.withzip.min`  
`MHC.shortread.zipcodes`
- ▶ CHM13 is a haplotype path (`chm13#0#ctg1#0`), not a second reference; K-PanRef and CPC haplotypes come from their own graphs, not from assemblies; JaSaPaGe NA18952 carries one haplotype twice
- ▶ Workflow and job scripts: [github.com/leechuck/pangenome-bh26](https://github.com/leechuck/pangenome-bh26) (`hla/nig/mhc_mc_graph.sbatch`); questions: [robert.hoehndorf@kaust.edu.sa](mailto:robert.hoehndorf@kaust.edu.sa)